

A Survey of Cooperative Multi-agent Reinforcement Learning for Multi-task Scenarios

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ABSTRACT

Cooperative multi-agent reinforcement learning (MARL) is a key technology for enabling cooperation in complex multi-agent systems. It has achieved remarkable progress in areas such as gaming, autonomous driving, and multi-robot control. Empowering cooperative MARL with multi-task decision-making capabilities is expected to further broaden its application scope. In multi-task scenarios, cooperative MARL algorithms need to address 3 types of multi-task problems: reward-related multi-task, arising from different reward functions; multi-domain multi-task, caused by differences in state and action spaces, state transition functions; and scalability-related multi-task, resulting from the dynamic variation in the number of agents. Most existing studies focus on scalability-related multi-task problems. However, with the increasing integration between large language models (LLMs) and multi-agent systems, a growing number of LLM-based multi-agent systems have emerged, enabling more complex multi-task cooperation. This paper provides a comprehensive review of the latest advances in this field. By combining multi-task reinforcement learning with cooperative MARL, we categorize and analyze the 3 major types of multi-task problems under multi-agent settings, offering more fine-grained classifications and summarizing key insights for each. In addition, we summarize commonly used benchmarks and discuss future directions of research in this area, which hold promise for further enhancing the multi-task cooperation capabilities of multi-agent systems and expanding their practical applications in the real world.

In recent years, the field of artificial intelligence has achieved significant progress due to the emergence of reinforcement learning (RL) techniques^[1]. As illustrated in Fig. 1a, these algorithms primarily investigate agent-environment interaction and decision-making, optimizing agent policies through trial-and-error methods, with the ultimate goal of solving various real-world decision-making problems. Since real-world decision problems often

involve interactions among multiple agents, and the coupling relationships between agents make single-agent RL algorithms ineffective for solving multi-agent decision problems (*e.g.*, esports, logistics allocation, intelligent transportation). Therefore, research on cooperative multi-agent RL (MARL) algorithms, as shown in Fig. 1b, holds tremendous value and will further expand the application scope of decision-making algorithms^[2].

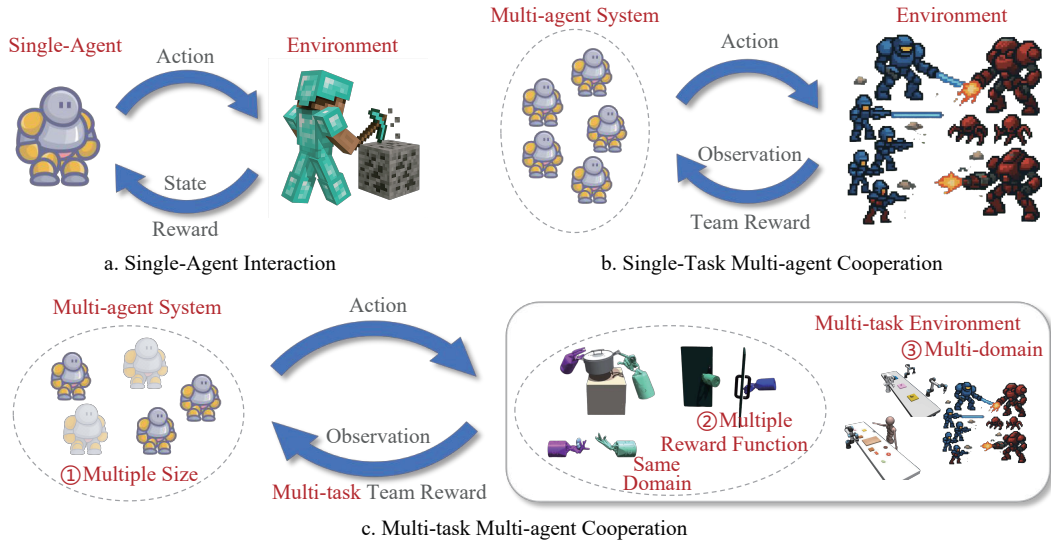
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Note: In Fig. 1a, a single agent interacts with the environment. In Fig. 1b, multiple agents are involved, and the goal is to maximize the team reward. In Fig. 1c, 3 types of multi-task problems are illustrated: ① represents varying numbers of agents, ② denotes different reward functions within the same task domain, and ③ indicates different task domains.

Fig. 1 Task Setting Comparison

However, whether in single-agent or MARL, these achievements in decision-making domains have been accomplished separately across different fields. Taking single-agent scenarios as an example, AlphaZero can be trained to surpass human performance in Go, Shogi, and Chess individually, but it cannot produce a single agent capable of defeating humans in all three board games simultaneously. The need to achieve such capability has driven research into multi-task cooperative MARL algorithms^[3]. As shown in Fig. 1c, in multi-agent cooperative tasks, the scale of multi-agent systems is often not static. It typically changes dynamically during the decision-making process, leading to variations in the number of agents, thereby presenting scalability-related multi-task challenges for cooperative MARL algorithms. Additionally, similar to single-agent scenarios, multi-task challenges in multi-agent domains include reward-related multi-task^[4], which involves adapting to different reward functions within the same task domain (e.g., running and jumping in robot control), and multi-domain multi-task^[5], where a single policy adapts to decision-making across different task domains with varying state and action spaces (e.g., robot control and board games like Go). Current multi-task cooperative MARL algorithms predominantly address scalability-related multi-task challenges, mainly focusing on adapting to the changes in the number of agents. However, a few methods have achieved multi-task cooperative decision-making through supervised learning. Furthermore, with the rise of large language models (LLMs), numerous studies have implemented multi-agent cooperation across multiple task domains using natural language.

Given the practical demands and rapid development of multi-agent multi-task RL, it is necessary to conduct a systematic review and summary of this

field. Unfortunately, as shown in Table 1, most existing surveys either focus on the challenges of cooperative training in multi-agent reinforcement learning^[6-7], organize current works from theoretical^[8-9] and algorithmic framework perspectives^[10], or introduce multi-task learning primarily in single-agent settings^[11]. Specifically, Ref.[11] surveys multi-task learning from the single-agent perspective; Refs.[6-10,12] focus on theoretical analyses and general cooperative MARL algorithms; and Refs.[13-17] review application-driven cooperative MARL benchmarks, including scenarios like autonomous driving and open-environment cooperation. Different from them, our work uniquely emphasizes multi-task cooperative settings, systematically summarizing algorithms, techniques, benchmarks, and future trends, thus filling an important gap between multi-task learning and cooperative MARL research.

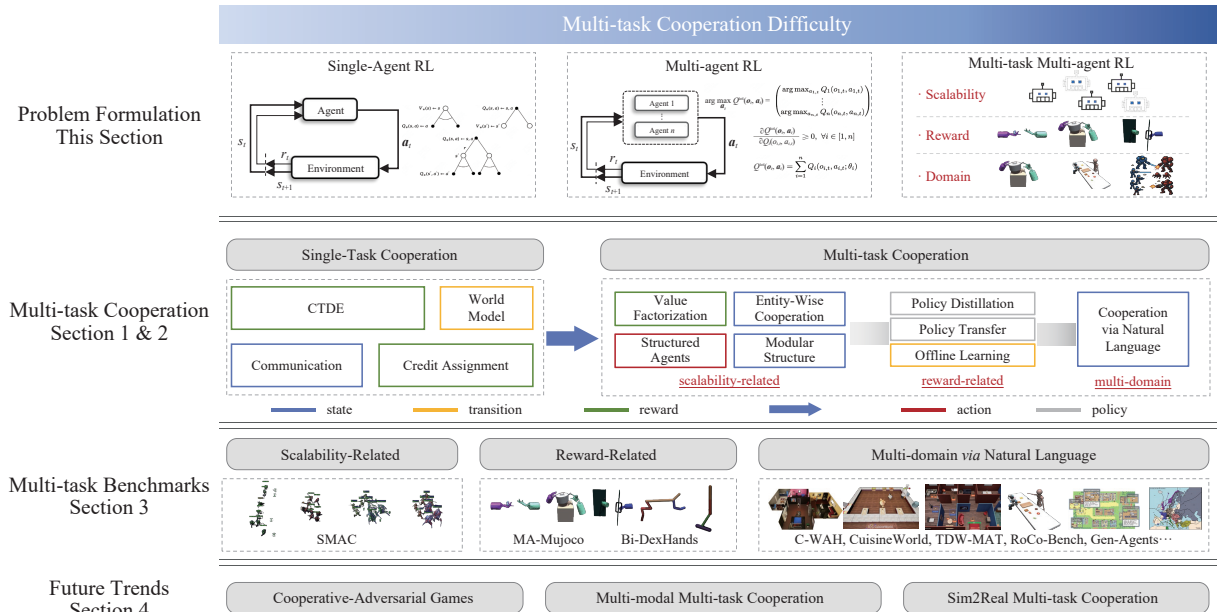
Table 1 Current Surveys in Multi-task RL and Cooperative MARL

Surveys	Multi-task	Cooperation	Benchmarks
Ref.[11]	√	×	×
Refs.[13-17]	×	√	√
Refs.[6-10,12]	×	√	×
Ours	√	√	√

As shown in Fig. 2, this paper focuses on cooperative MARL algorithms and systematically reviews cooperative MARL approaches for multi-task scenarios. Firstly, this section introduces the relevant background knowledge of RL and MARL, primarily covering the theoretical and algorithmic foundations required for cooperative training, which are essential prerequisites for achieving cooperative decision-making in multi-task settings. In Fig. 2, we annotate the existing technical approaches on the color-coded boxes based on the 5 key elements of RL: state,

transition, reward, action, and policy. The primary objective in multi-task cooperative MARL research remains endowing multi-agent systems with effective cooperative decision-making capabilities. Secondly, Section 1 introduces the current research landscape in traditional cooperative MARL. A series of key techniques employed in this field help address subsequent challenges in multi-task cooperative MARL, including centralized training with decentralized execution (CTDE) frameworks, inter-agent communication, credit assignment, and world models. Thirdly, Section 2 focuses on multi-task MARL algorithms. To address more complex multi-task cooperation problems, these research efforts have conducted a more detailed investigation into each element. Section 2 first clarifies the definition of multi-task settings in multi-agent systems, including scalability-related multi-task, reward-related multi-task, and cross-domain multi-task

scenarios. Currently, most research concentrates on scalability-related multi-task problems in cooperative multi-agent settings, introducing technical approaches such as entity-wise methods, structured agents, and scalable value decomposition. It also covers recent work leveraging the generalizability of LLMs for reward-related multi-task and cross-domain multi-task applications. Next, Section 3 introduces mainstream benchmarks in multi-agent scenarios and explains how existing research constructs multi-task settings based on these benchmarks. Future researchers can select suitable benchmarks for testing. Finally, Section 4 discusses future directions for cooperative MARL algorithms in multi-task scenarios, including multi-task decision-making in cooperative-competitive games, multi-modal information-aided multi-task cooperation, and multi-task cooperation in more complex Sim2Real (simulation-to-reality) environments.



Note: This section provides problem definitions and background knowledge on RL, cooperative MARL, and multi-task cooperation. Sections 1 and 2 summarize existing research on cooperative MARL algorithms in single-task and multi-task scenarios, respectively, categorized based on their technical approaches. Section 3 introduces the benchmarks commonly used in the three types of multi-task settings, while Section 4 discusses future directions in this field. SMAC: StarCraft multi-agent challenge; MA-Mujoco: Multi-agent Mujoco; Bi-DexHands: Bimanual dexterous hands; C-WAH: Collaborative watch-and-help; CuisineWorld: Cuisine collaboration world; TDW-MAT: Three-dimensional world multi-agent transport; RoCo-Bench: Multi-robot collaboration benchmark; Gen-Agents: Generalizable agents.

Fig. 2 Structure of the Paper

The main contributions of this study can be summarized as follows:

1) We exclusively investigate the cooperative MARL algorithms for multi-task scenarios. By progressively constructing the problem formulation from RL to cooperative MARL and ultimately to multi-task cooperative MARL, we analyze the theoretical and technical challenges in multi-task decision-making for multi-agent systems.

2) We provide a thorough survey of the latest advancements in this field, integrating insights from

both multi-task RL and cooperative MARL. We categorize multi-task problems in multi-agent settings into three major classes, with sub-categories and insights for each. Additionally, we compile commonly used benchmarks in current research.

3) We identify the limitations of existing approaches in the field and propose future research directions to address these challenges. These recommendations aim to enhance the multi-task cooperative capabilities of multi-agent systems, thereby expanding their potential for real-world applications.

Notations $\mathbb{S}$: state space; $\mathbb{A}$: action space; $\mathbb{T}$: state transition distribution function; $\mathbb{O}$: joint observation space; R : reward function; γ : discount factor; s : state; a : action; π : policy; s_{t+1} : state at t ; a_t : action at t ; r_t : reward at t ; G_t : accumulated discounted reward at t ; V_π : state value; Q_π : state-action value; s' : state at next timestep; $a_{i,t}$: action of agent i at time t ; $o_{i,t}$: observation of agent i at time t ; $\mathbf{a}_t$: joint action at time t ; $\mathbf{o}_t$: joint observation at time t ; $Q_{i,t}$: action value of agent i at time t ; Q_t^{tot} : joint action value at time t .

1 Background

1.1 RL Algorithms

RL algorithms are considered an important technological path to achieving general artificial intelligence. They optimize agent policies through a “trial-and-error” approach in response to reward signals obtained from interactions between the agent and the environment. The optimization objective is to maximize the expected cumulative discounted reward for each state. The interaction process can be modeled as a Markov decision process (MDP), typically represented as a tuple $(\mathbb{S}, \mathbb{A}, \mathbb{T}, R, \gamma)$, where each component is defined as follows:

1) State Space $\mathbb{S}$: The range of values for a state $s \in \mathbb{S}$. The state s describes all information about the environment at the current time step, such as the board configuration in Go or the robot's pose and velocity in a robotic control task.

2) Action Space $\mathbb{A}$: The range of values for an action $a \in \mathbb{A}$. The agent chooses an action $a \sim \pi(\cdot | s)$ based on the state s at any given time, where π is the agent policy used for decision-making. There are two types of action spaces, continuous and discrete. For example, in the movement of the agent in a plane, discrete action models it as four actions: up, down, left, right; while continuous action requires specifying a continuous vector as the direction of movement.

3) State Transition Distribution Function $\mathbb{T}$ ($\mathbb{S} \times \mathbb{A} \rightarrow \mathbb{S}$): At time step t , the environment transitions from state s_t to a new state $s_{t+1} \sim \mathbb{T}(\cdot | s_t, a_t)$ based on the action a_t provided by the agent, requiring the transition distribution $\mathbb{T}$ to maintain time invariance.

4) Reward Function R ($\mathbb{S} \times \mathbb{A} \rightarrow \mathbb{R}$): The real number set $\mathbb{R}$ represents the range of reward signals. After executing action a_t in state s_t , the environment provides a reward signal $r_t = R(s_t, a_t)$ to evaluate the quality of the agent's decision behavior.

5) Discount Factor $\gamma \in [0, 1]$: Used to calculate accumulated discounted reward $G_t = \sum_{j=0}^{\infty} \gamma^j r_{t+j}$ (also known as return), affecting the contribution of future rewards in the decision trajectory. The larger γ indicates more emphasis on long-term rewards, and vice versa for short-term rewards.

RL studies sequential decision-making problems in which the agent's state evolves over time and must adhere to the Markov property. Due to this Markov property, the optimization of RL differs significantly from that of supervised learning. The agent often requires multiple rounds of interaction with the environment to receive a series of reward signals, and the impact of decision actions at a particular time may take a long time to become apparent, especially in complex decision problems like Go with sparse reward signals. Therefore, the optimization of the agent's policy must enhance the entire trajectory's return based on current interaction data to achieve long-term goals. In contrast, the impact of outputs in supervised learning is instantaneous and affects only one time step with immediate feedback on correctness (*i.e.*, prediction error), accompanied by a well-defined labeled training dataset, which RL lacks.

Following the MDP model for solving tasks, two key functions are defined in RL to describe the expected return of the agent policy π in any state. First of all, the state value function $V_\pi(s_t)$ is defined as the expected cumulative discounted reward starting from the state s_t at time t , up to either the termination time T or infinitely, with the policy π unchanged.

$$V_\pi(s_t) = \mathbb{E}_\pi[G_t | s_t] = \mathbb{E}_{a \sim \pi(\cdot | s)} \left[\sum_{j=0}^{\infty} \gamma^j r_{t+j} | s_t \right] \quad (1)$$

The expectation operator $\mathbb{E}$ indicates that all actions a are sampled according to policy π . Similarly, the state-action value function $Q_\pi(s_t, a_t)$ represents the expected cumulative discounted reward that the agent can obtain starting from state s_t and taking action a_t with policy π unchanged.

$$Q_\pi(s_t, a_t) = \mathbb{E}_\pi[G_t | s_t, a_t] = \mathbb{E}_\pi \left[\sum_{j=0}^{\infty} \gamma^j r_{t+j} | s_t, a_t \right] \quad (2)$$

The most important formula in RL is the Bellman Equation, which denotes the relationship between the V-function and Q-function.

$$\begin{aligned} V_\pi(s_t) &= \mathbb{E}_\pi[r_t + \gamma V_\pi(s_{t+1})] \\ Q_\pi(s_t, a_t) &= \mathbb{E}_\pi[r_t + \gamma Q_\pi(s_{t+1}, a_{t+1})] \end{aligned} \quad (3)$$

In the Q-function Bellman equation, the action at time $t+1$ is sampled according to π , $a_{t+1} \sim \pi(\cdot | s_{t+1})$. Furthermore, there exists a certain relationship between the Q-function and the V-function: the state value of an agent at state s_t equals the expectation of the Q-value corresponding to the sampled action according to policy π .

$$V_\pi(s_t) = \sum_{a_t \in \mathbb{A}} \pi(a_t | s_t) Q_\pi(s_t, a_t) \quad (4)$$

The Q-value corresponding to the agent state-action pair (s_t, a_t) is equivalent to the expectation value of the V-value corresponding to the next state

of environmental transition.

$$Q_\pi(s_t, a_t) = R(s_t, a_t) + \gamma \sum_{s' \in \mathbb{S}} \mathbb{T}(s' | s_t, a_t) V_\pi(s') \quad (5)$$

The goal of RL is to obtain an agent policy that maximizes the expected return across all states. With the optimal V-function defined as $V^*(s_t) = \max_\pi V_\pi(s_t)$, $s_t \in \mathbb{S}$ and the optimal Q-function as $Q^*(s_t, a_t) = \max_\pi Q_\pi(s_t, a_t)$, $s_t \in \mathbb{S}, a_t \in \mathbb{A}$. The optimal Bellman Equation links optimal Q-functions and optimal V-functions, allowing the optimal policy $\pi^*(s) = \arg \max_a Q^*(s, a)$ to be determined through optimal Q-functions. It also possesses the optimization property^[18] that regardless of the initial state and decision action, the remaining steps still form the optimal policy for the remaining problem. Simple tasks allow the agent's optimal policy to be solved using dynamic programming methods, but more complex problems require subsequent deep RL algorithms.

1.2 Decentralized Partially Observable MDP

Cooperative MARL systems consider a fully cooperative multi-agent system in partially observable environments, where each agent independently obtains its local observation information from the environment and makes decisions accordingly. This problem is formally defined as a decentralized partially observable MDP (Dec-POMDP), represented by the tuple $\{n, \mathbb{S}, \mathbb{A}, \mathbb{T}, \mathbb{O}, R, \gamma\}$, where n is the number of agents in the cooperative multi-agent system, $\mathbb{A}$ and $\mathbb{O}$ are the joint action space and joint observation space respectively, and $\mathbb{S}$ is the global state space of the entire multi-agent system. From the perspective of the multi-agent system, the joint policy π takes joint actions $\mathbf{a}_t = \{a_{1,t}, \dots, a_{n,t}\} \sim \pi(\cdot | \mathbf{o}_t)$ based on joint observations $\mathbf{o}_t = \{o_{1,t}, \dots, o_{n,t}\}$. However, in the decentralized setting, each agent makes decisions individually based on its local observation $a_{i,t} \sim \pi_i(\cdot | o_{i,t})$, where π_i denotes the individual policy of agent i . The environment then provides a shared team reward signal r_t according to the function $R: \mathbb{S} \times \mathbb{A} \rightarrow \mathbb{R}$ and transitions to the next global state $s_{t+1} \in \mathbb{S}$ via the dynamics function $\mathbb{T}(s_{t+1} | s_t, \mathbf{a}_t)$. Finally, γ represents the discount factor for the cumulative discounted reward G_t in RL.

In Dec-POMDP, the cooperative multi-agent system shares the same team reward function. Therefore, the global state value function $V_\pi(s_t)$ is defined as the expected discounted cumulative reward starting from state s , while the Q-function additionally conditions on the joint action $\mathbf{a}_t$, representing the expected cumulative discounted reward after taking joint action $\mathbf{a}_t$ in state s_t .

$$\begin{aligned} V_\pi(s) &= \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = s \right] \\ Q_\pi(s, \mathbf{a}) &= \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = s, \mathbf{a}_t = \mathbf{a} \right] \end{aligned} \quad (6)$$

The advantage function is defined as the difference between the Q-function and V-function: $A_\pi(s, \mathbf{a}) = Q_\pi(s, \mathbf{a}) - V_\pi(s)$. Under the joint policy π , the expectation of the advantage function equals zero.

$$\mathbb{E}_{\mathbf{a}_t \sim \pi(\cdot | s_t)} A_\pi(s_t, \mathbf{a}_t) = 0 \quad (7)$$

Compared to the Q-function, the advantage function can evaluate joint actions $\mathbf{a}$ with lower variance. To solve Dec-POMDP problems, the most popular MARL framework is CTDE, which treats the multi-agent system as a whole during training to avoid non-stationarity issues in decentralized training, while during execution each agent makes decisions based only on its local observations. This effectively reduces the complexity of optimal decision-making in multi-agent systems while satisfying the partial observability requirement.

1.3 Basis of Multi-agent Cooperation

With the optimization objective of maximizing the global joint Q-value of the multi-agent system, Q-mixing^[19] (QMIX) and related improved methods require the algorithm to satisfy the individual global max (IGM) condition, ensuring that the optimization objectives of all agents are consistent with the global optimization objective of the entire multi-agent system, thereby guaranteeing the overall cooperation of the multi-agent system.

Definition 1 For a multi-agent system with n agents, if the optimal joint action is equivalent to the set of agent actions that maximize their individual action-value functions, then the system satisfies the IGM condition. This can be expressed as:

$$\arg \max_{\mathbf{a}_t} Q^{\text{tot}}(\mathbf{o}_t, \mathbf{a}_t) = \begin{pmatrix} \arg \max_{a_{1,t}} Q_1(o_{1,t}, a_{1,t}) \\ \vdots \\ \arg \max_{a_{n,t}} Q_n(o_{n,t}, a_{n,t}) \end{pmatrix} \quad (8)$$

Since it is difficult to verify whether the algorithm satisfies the IGM condition in practice, the following monotonicity condition is often used to ensure that the optimization direction of all agents is consistent with the global optimization direction of the multi-agent system.

$$\frac{\partial Q^{\text{tot}}(\mathbf{o}_t, \mathbf{a}_t)}{\partial Q_i(o_{i,t}, a_{i,t})} \geq 0 \quad \forall i \in [1, n] \quad (9)$$

If $Q^{\text{tot}}(\mathbf{o}_t, \mathbf{a}_t)$ is monotonically factorized in the form of Eq. (9), then this factorization satisfies the IGM condition, meaning that maximizing the joint action-value is equivalent to maximizing each agent's action-value.

1.4 Multi-agent Cooperation for Multi-task Scenarios

In the single-task multi-agent cooperation, the MDP of the multi-agent system is unchanged. In multi-task scenarios, it is necessary to maintain a finite task set $\mathcal{T} = \{\mathcal{M}_k\}$, each task is characterized

by a Dec-POMDP, denoted as $\mathcal{M}_k$. Thus, the optimization objective of multi-task MARL can be defined as:

$$J(\pi) = \max_{\pi} \mathbb{E}_{\mathcal{M}_k \in \mathcal{T}} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{R}_k(s_t, \mathbf{a}_t) \right] \quad (10)$$

In the multi-agent setting, the multi-task encompasses three sub-concepts:

1) Scalability-Related Multi-task: Multi-task scenarios brought about by dynamic changes in the number of agents in multi-agent systems, where different tasks in $\mathcal{T}$ differ in the number of agents n .

2) Reward-Related Multi-task: Multi-task scenarios resulting from differences in reward functions, where different tasks in $\mathcal{T}$ differ in the reward function R , such as running and jumping belonging to different tasks.

3) Multi-domain Multi-task: Apart from differences in reward functions, this type of multi-task scenario also requires that different tasks in $\mathcal{T}$ have differences in the state space $\mathbb{S}$, action space $\mathbb{A}$, and dynamics function $\mathbb{T}$.

2 Cooperative MARL for Single-Task Scenarios

Since cooperation is still one of the core aspects of multi-task multi-agent cooperation, this section first introduces some traditional paradigms of single-task MARL, including Traditional CTDE-based MARL, communication-based MARL, implicit and explicit credit assignment, and model-based MARL (MBMARL) methods. Table 2 summarizes cooperative multi-agent reinforcement learning methods.

Table 2 A Review of Cooperative MARL Methods in Single-Task Scenarios

Paradigm	Motivation	Representative Methods
Traditional CTDE-Based MARL	Non-stationary and huge state & action space for centralized decision-making	VDN ^[20] , QMIX ^[19] , QTRAN ^[21] , QPLEX ^[22] , VDAC ^[23] , CPEG ^[24]
Communication-Based MARL	Effective cooperation in the partially observable multi-agent setting	RIAL & DIAL ^[25] , CommNet ^[26] , MARGIN ^[27] , MASIA ^[28] , I2C ^[29] , CCOMA ^[30] , NCC ^[31] , GraphComm ^[32] , ETCNet ^[33] , SchedNet ^[34] , IC3Net ^[35]
Explicit Credit Assignment	Team reward allocation for cooperative joint policy updates	COMA ^[36] , LI <i>et al.</i> ^[37] , SQDDPG ^[38] , SHAQ ^[39] , DAE ^[40] , HAN <i>et al.</i> ^[41]
MBMARL	Improve the sample efficiency for MARL methods	MAMBPO ^[42] , MBOM ^[43] , CPS ^[44] , MDBS ^[45] , MAMBA ^[46] , MAG ^[47] , MACD ^[48]

Note: VDN: Value decomposition network; QTRAN: Q-transformation; QPLEX: DuPlex dueling Q; VDAC: Value-decomposition actor-critic; CPEG: Consistency policy with intention guidance; RIAL: Reinforced inter-agent learning; DIAL: Differentiable inter-agent learning; CommNet: Communication network; MARGIN: Multi-agent reinforcement learning with graphical mutual information maximization; MASIA: Multi-agent communication via self-supervised information aggregation; I2C: Individually inferred communication; CCOMA: Graph convolutional counterfactual multi-agent policy gradient; NCC: Neighborhood cognition consistent; GraphComm: Graph-based communication network; ETCNet: Event-triggered communication network; SchedNet: Scheduled communication network; IC3Net: Individualized controlled continuous communication model; COMA: Collaborative counterfactual multi-agent policy gradient; SQDDPG: Shapley Q DDPG; SHAQ: Shapley Q-learning; DAE: Difference advantage estimation; MAMBPO: Multi-agent model-based policy optimization; MBOM: Model-based opponent modeling; CPS: Cooperative prioritized sweeping; MDBS: Model-based dynamic shielding; MAMBA: Multi-agent model-based approach; MAG: Models as agents; MACD: Multi-agent counterfactual dreamer.

2.1 Traditional CTDE-Based MARL

CTDE is an important branch of cooperative MARL algorithms, which simultaneously addresses the key challenges in the MARL training process. As shown in Fig. 3, it treats the multi-agent system as a whole during centralized training to avoid the issue of environmental non-stationarity and ensure the uniqueness and stability of the state transition process. During actual decision-making, the agents only output actions based on their local observations for decentralized execution, thereby alleviating the exponentially growing decision complexity^[49].

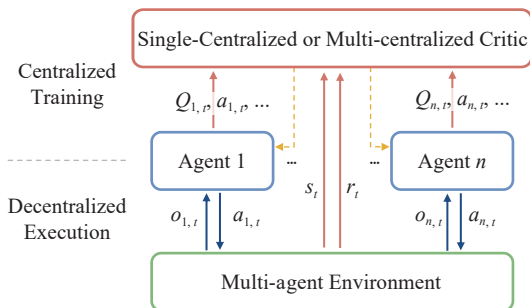


Fig. 3 Illustration of the QMIX Algorithm Framework^[19]

VDN^[20] and QMIX^[19] are the most classic CTDE algorithms. VDN first uses the sum of all agents' Q-functions as the joint Q-function of the entire system for centralized training. As shown in Fig. 4, QMIX further introduces the concept of a mixing network, which maps all agents' Q-values to the joint Q-value, effectively enhancing the efficiency of centralized training. Subsequent research has largely followed the QMIX framework, focusing on modifying the mixing network to reduce the complexity of cooperative training and improve training stability. QTRAN^[21] decomposes the joint Q-value by designing a specialized learning objective, transforming the original joint Q-function into a new, easily decomposable function family. QPLEX^[22] proposes a novel duplex dueling network structure to represent the joint Q-function of the cooperative system, reducing the training difficulty of the mixing network by introducing advantage value functions. CPEG^[24] enhances the exploration and cooperation in cooperative MARL by using a multimodal policy for exploration and sharing consensus for coordination among agents. VDAC^[23] further extends the

decomposition of the joint Q-value to the joint V-value, enabling QMIX-like algorithms to more flexibly utilize advantage value functions for cooperative training under the actor-critic reinforcement learning framework. However, these algorithms all follow QMIX's fixed-dimensional design for the

mixing network, failing to account for the dynamic nature of the multi-agent system's cooperative process, which leads to a decline in the mixing network's ability to approximate the joint Q-function and limits the system's cooperative capability.

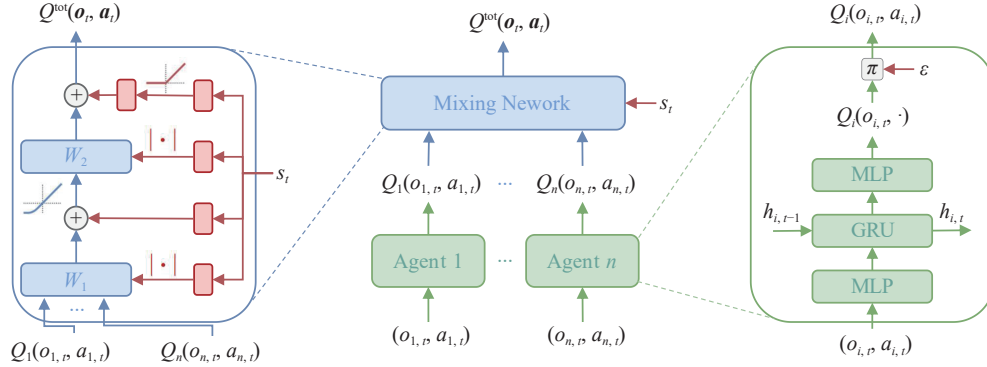


Fig. 4 Illustration of the CTDE Framework

2.2 Communication-Based MARL

Although QMIX-based algorithms have achieved outstanding performance in many multi-agent cooperative tasks, they may still struggle to make effective decisions during decentralized execution due to insufficient observational information. To address this, some algorithms have proposed communication-based MARL approaches under the partially observable multi-agent system framework, enhancing individual agents' perception and decision-making capabilities by introducing information exchange mechanisms among agents^[50]. A subset of communication-based MARL algorithms still adopts the CTDE framework. RIAL and DIAL^[25], as early

works in this category, establish full communication channels between agents as shown in Fig. 5a, effectively creating a quasi-centralized decision-making framework. In this approach, every pair of agents in the system is connected by a communication channel, meaning each agent's decision-making must account for information from all other agents, which clearly hampers scalability in complex tasks. CommNet^[26] improves scalability to some extent by averaging communication messages before transmission. Ref.[51] employs graph neural networks (GNNs) to compute the connections between agents, primarily using a graph convolutional network module to achieve information exchange among agents in a multi-agent system.

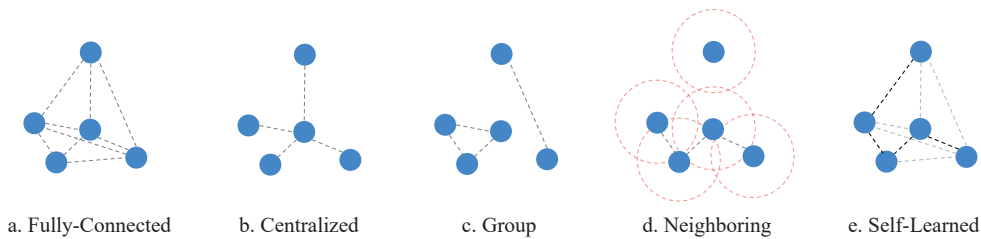


Fig. 5 The Classification of Communication Structures in Communication-Based MARL Algorithms

More recent algorithms focus on reducing communication overhead by optimizing the multi-agent communication topology, allowing agents to communicate only when necessary to alleviate information processing burdens. CCOMA^[30] adopts a centralized communication protocol in Fig. 5b. Ref.[52] proposes a group communication mechanism in Fig. 5c where agents autonomously decide whether to act as initiators or receivers within a communication group. As a result, the system contains only a limited amount of communication information, confined within the scope of the

communication group. This approach of group-based communication and decision-making effectively enhances system collaboration. As shown in Fig. 5d, MARGIN^[27] employs GNNs to aggregate and process neighboring agents' information, optimizing agents' perception of neighbor features and topological relationships *via* mutual information maximization. MASIA^[28] introduces a self-supervised training paradigm to enhance the representation power of agent communication protocols. Communication Transformer^[53] models multi-agent communication as a learnable graph and jointly

optimizes the topology and parameters through a bi-level process like Fig. 5e. I2C^[29] leverages a pre-trained network to determine communication partners, further improving performance from a topological perspective. Besides, NCC^[31] and GraphComm^[32] adopt the CTDE framework while implementing agent communication based on different rule-based topologies.

Other approaches tackle more realistic communication challenges. ETCNet^[33] proposes an event-triggered communication mechanism to reduce bandwidth consumption and improve efficiency. SchedNet^[34] and IC3Net^[35] employ gating modules to filter out unnecessary communications, thereby reducing frequency. While these communication-based algorithms successfully solve cooperative multi-agent problems under the CTDE framework, their reliance on centralized training, particularly the need for global state information, severely limits their performance in complex scenarios (*e.g.*, large-scale systems).

2.3 Credit Assignment for Cooperation

In cooperative multi-agent systems, since all agents share a common team reward, credit assignment is necessary to evaluate individual contributions. This concept is widely applied in cooperative MARL. Methods represented by QMIX^[19] and its numerous variants (*e.g.*, Refs. [54-56]) adopt implicit credit assignment, where agents' policies are updated based on gradients from the joint Q-function during centralized training. While simple and easy to implement, this approach heavily relies on the accuracy of the joint Q-function and suffers from inefficiency in training.

Another category of methods employs explicit credit assignment, using counterfactual reasoning or Shapley theory^[37] to compute each agent's contribution and directly derive individual rewards from the team reward. For instance, COMA^[36] marginalizes an agent's action using the joint value function to compute a counterfactual advantage for contribution assessment. SQDDPG^[38] introduces Shapley Q-values in the critic module of DDPG to distribute team rewards. SHAQ^[39] derives the Shapley-Bellman operator, addressing SQDDPG's theoretical convergence limitations. Furthermore, LI *et al.*^[57] propose a nucleolus-based credit assignment that ensures convergent reward distribution by minimizing maximum dissatisfaction in multi-agent coalitions.

However, since value functions for credit assignment remain accurate only near the current policy distribution, model-free RL algorithms struggle to precisely evaluate contributions. In contrast, DAE^[40] constructs a reward predictor to reconstruct global rewards into difference rewards based on potential functions. HAN *et al.*^[41] propose a single-step state predictor, computing Shapley advantage values to

guide policy updates. However, these methods depend on state-value functions, which may yield inaccurate estimates outside the current policy distribution, leading to unreliable credit assignment.

2.4 MBMARL

Model-based RL (MBRL) algorithms utilize real-world data obtained from the interaction between an agent's policy and the environment to train a world model. This world model approximates the dynamics function, reward function, and other aspects of the real environment, and then uses virtual interaction data between the agent's policy and the world model to train the agent. In single-agent scenarios, the most renowned MBRL algorithm is the Dreamer series^[58-60]. Its main contribution lies in constructing a high-performance world model framework, which optimizes the agent's policy based on virtual interaction data generated by this world model, effectively improving the algorithm's sample efficiency.

With the advancement of MARL algorithms, some methods have begun to introduce MBRL into multi-agent problems. For example, MAMBPO proposes a MBMARL approach, achieving outstanding sample efficiency in multi-robot tasks^[42]. MBOM considers opponent policy optimization and leverages the environment model to obtain the best response from opponents^[43]. Additionally, CPS employs a cooperation graph to identify the most urgently needed state-action pairs for updates, effectively addressing cooperation challenges in large-scale discrete environments^[44]. MDBS introduces a model-based approach to monitor and correct unsafe behaviors in MARL agents. These works adopt a naive framework for constructing their world models, resulting in inefficient virtual data generation processes^[45].

MAMBA extends Dreamer from single-agent to multi-agent settings, leveraging Dreamer's long-term prediction capability in policy training to achieve superior sample efficiency compared to model-free MARL methods^[46]. MAG further enhances MAMBA's performance by treating virtual trajectory generation as a multi-step decision-making process, mitigating cumulative errors in virtual data^[47]. However, their fully decentralized multi-agent world models lead to inaccurate predictions of the global state transition process of the system, thereby resulting in inaccurate virtual data describing the system's global state transitions. MACD^[48] proposes a centralized world model to achieve higher multi-agent prediction accuracy, significantly improving the quality of virtual trajectory generation. It also introduces a novel explicit credit assignment mechanism based on counterfactual advantage values, effectively reducing the difficulty of cooperative training in multi-task scenarios. Subsequently, both MABL^[61] and global-aware world model^[62] leverage global infor-

mation during training to enhance representation learning and improve multi-agent policy performance under decentralized execution.

3 Cooperative MARL for Multi-task Scenarios

Currently, most existing multi-task cooperative MARL algorithms remain at the level of scalability-related multi-task approaches, primarily addressing challenges arising from dynamic changes in the number of agents due to system dynamics. A few methods have achieved cooperative decision-making for multiple tasks through supervised learning^[63-64]. Additionally, with the rise of LLMs, numerous

studies have explored multi-agent cooperation in multi-task domains using natural language, though most of these methods do not incorporate RL for optimal decision-making.

As shown in Fig. 2, this section starts with scalability-related multi-task cooperation, introducing three technical approaches: scalable value factorization and structured agents, as well as entity-wise cooperation. Next, this section introduces three technical branches based on policy distillation, policy transfer, and offline MARL. Finally, we focus on multi-domain multi-task multi-agent cooperation based on natural language. Table 3 provides a summary review of cooperative MARL in multi-Task scenarios.

Table 3 A Review of Cooperative MARL Methods in Multi-task Scenarios

Paradigm	Motivation	Representative Methods
Scalable Value Factorization	Adaptation to time-varying multi-agent system size in scalability-related multi-task scenarios	Qatten ^[65] , ROMA ^[66] , RODE ^[67] , NVIF ^[50] , WQMIX ^[54] , MG2L ^[68] , MAT ^[56] , TransQMIX ^[55]
Structured Agents	Adaptation to time-varying agent action spaces in scalability-related multi-task scenarios	MAAC ^[69] , UNMAS ^[70] , ASN ^[71] , AMARL ^[72] , LDR ^[73]
Entity-Wise Cooperative MARL	Adaptation to both issues in scalability-related multi-task scenarios	REFIL ^[74] , PTM ^[75] , HUANG <i>et al.</i> ^[76] , OPT ^[77] , ALMA ^[78] , UPDeT ^[79] , TETQmix ^[80]
Policy Distillation	Cooperation consistency of agent policies across varying multi-agent system sizes	OMIDSHAFIE <i>et al.</i> ^[81] , SAML ^[82]
Policy Transfer	Same as above	MATTER ^[83] , MRA ^[84] , PAT ^[85] , MATE ^[86] , VACL ^[87]
Offline Methods	Same as above	MAGENTA ^[88] , ODIS ^[63] , INS ^[89]
Modular Structure	Adaptation to varying reward functions across tasks	SMP ^[90] , M3 ^[91] , ADMN ^[92] , CSAOT ^[93] , YANHUI ^[94]
Natural Language-Based	Cooperation in multi-domain multi-task scenarios	GUO <i>et al.</i> ^[95] , Richelieu ^[96] , TextStarCraft-II ^[97] , MAGIS ^[98] , CHEN <i>et al.</i> ^[99] , AutoGen ^[100] , Meta-GPT ^[101] , CAMEL ^[102] , MAD ^[103] , LI <i>et al.</i> ^[104] , GroupDebate ^[105] , ChatEval ^[106] , RoCo ^[107] , MLAQ ^[108] , CoELA ^[109] , LI <i>et al.</i> ^[110] , ProAgent ^[111]

Note: Qatten: Q-attention; ROMA: Role-oriented MARL; RODE: Roles to decompose multi-agent tasks; NVIF: Neighboring variational information flow; WQMIX: Weighted QMIX; MG2L: Meta MARL *via* global-to-local; MAT: Multi-agent Transformer; TransQMIX: Transformer-Q-Mixer; MAAC: Multi-agent actor-critic; UNMAS: Unshaped networks for multi-agent systems; ASN: Action semantics network; AMARL: Attention-based multi-agent RL; LDR: Learning discrete representation; REFIL: Randomized entity-wise factorization for imagined learning; PTM: Prioritized tasks mining; OPT: Interaction pattern disentangling; ALMA: Allocator-actor multi-agent architecture; UPDeT: Universal policy decoupling Transformer; TETQmix: Task-entity Transformers QMIX; SAML: Self-adaptive meta-learning algorithm; MATTER: Multi-agent transfer reinforcement learning *via* modelling task relationship; MRA: Learning meta representations for agents; PAT: Parallel knowledge transfer; MATE: Multi-agent task embedding; VACL: Variational automatic curriculum learning; MAGENTA: Multi-agent multi-game entity Transformer; ODIS: Offline coordination skill discovery; INS: Interaction-aware synthesis; SMP: Shared modular policies; M3: Modularization for multi-task and multi-agent; ADMN: Agent driven modular network; CSAOT: Cooperative multi-agent system for active object tracking; MAGIS: Multi-agent framework for GitHub issue resolution; Meta-GPT: Meta-generative pre-trained Transformer; CAMEL: Communicative agents for “Mind” exploration of large language model society; MAD: Multi-agent debate; MLAQ: Model-based LLM agent with Q-learning; CoELA: Cooperative embodied language agent.

3.1 Scalable Value Factorization

To address the scalability limitations of traditional CTDE algorithms, some works focus on the approximation capability of mixing networks under varying numbers of agents through network architectures. The Qatten algorithm^[65] introduces a multi-head attention mechanism, proposing a more general joint Q-function decomposition approach from both theoretical and practical perspectives, leveraging the flexibility of attention mechanisms to enhance the scalability of mixing networks. The ROMA^[66] and RODE^[67] algorithms build upon the

QMIX framework to develop MARL algorithms for role assignment, improving problem-solving capabilities in complex scenarios by assigning identical role features to agents with similar states. NVIF^[50] trains a communication model based on its pre-training stage to provide it with efficient communication capability among different tasks. WQMIX^[54], proposed by the original authors of QMIX, introduces a weighted QMIX operator to ensure approximation of arbitrary joint Q-functions. MG2L^[68] extends the CTDE to the multi-agent meta MARL and introduces a multilevel task encoder for joint global and local task inference. TransQMIX^[55]

comprehensively improves the QMIX algorithm based on Transformer models, achieving Transformer-based sequential decision-making in entity-oriented frameworks. Recently, Ref.[112] proposes a solution to permutation issues in multi-agent collaboration and Ref.[113] introduces a hierarchical consensus mechanism for multi-agent cooperation, while Ref.[114] explores a consensus-based MARL framework.

3.2 Structured Agents

The issue of varying agent action spaces has long existed in single-agent RL research^[115], and this problem becomes more pronounced in multi-agent scenarios due to changing numbers of agents^[116]. Some research approaches focus on agent networks, introducing human expertise to construct structured agent networks, with core contributions centered on technologies like multi-head attention (MHA)^[117] to aggregate input observation information. MAAC^[69] is the first to incorporate MHA modules into agent Q-networks to aggregate information from all agents. Adopting the CTDE framework, it extends the soft actor-critic algorithm^[118] from single-agent RL to multi-agent domains, enabling agents to develop scalability-related multi-task decision-making capabilities during training. UNMAS^[70] adopts a self-weighted Q mixing network to construct a scalable system joint Q-function, and divides the agent's action space on its functionality to adapt to the dynamic changes in the action space. ASN^[71] collects local observation information from agents and considers action semantics between agents to compute Q-values for corresponding actions, dynamically adjusting agent inputs and outputs based on changes in multi-agent systems. AMARL^[72] also adopts a similar approach to reconstruct agent networks, but with more standardized input-output

model constructions to further reduce training difficulty. BREDELL *et al.*^[119] expand the action space by introducing conventions as special cooperative action sequences, requiring agents to actively opt in for them to reach fruition. LDR^[73] enhances the robustness of MARL against observation noise by employing quantized discrete representations. By processing these representations through a specialized combiner for decision-making, LDR shows improved performance in noisy environments while maintaining efficiency in clean observations.

3.3 Entity-Wise Cooperative MARL Methods

REFIL^[74] proposes an entity-wise decision-making and training framework tailored for multi-agent systems with dynamically changing numbers of agents. It refers to all controllable and uncontrollable units in the environment as “entities”, where controllable entities are still called agents. Under this framework, the observation space of agents can vary with the number of observable entities, enhancing the flexibility of algorithm design. In implementation, REFIL builds upon the QMIX algorithm, adapting its agent network and mixing network to the entity-wise framework. As shown in Fig. 6, in the agent network, REFIL employs an MHA module to aggregate information from observable entities for decision-making, enabling adaptation to changes in observation space due to varying entity counts. For the centralized training mixing network, REFIL posits that similar subsystem structures should exist across different tasks. Thus, it introduces the concept of imagination space training on top of QMIX, where entities are randomly grouped in the imagination space to allow agents to train on diverse subsystems. This approach equips agents with decision-making capabilities in unseen scenarios, where “scenarios” refer to variations in entity counts.

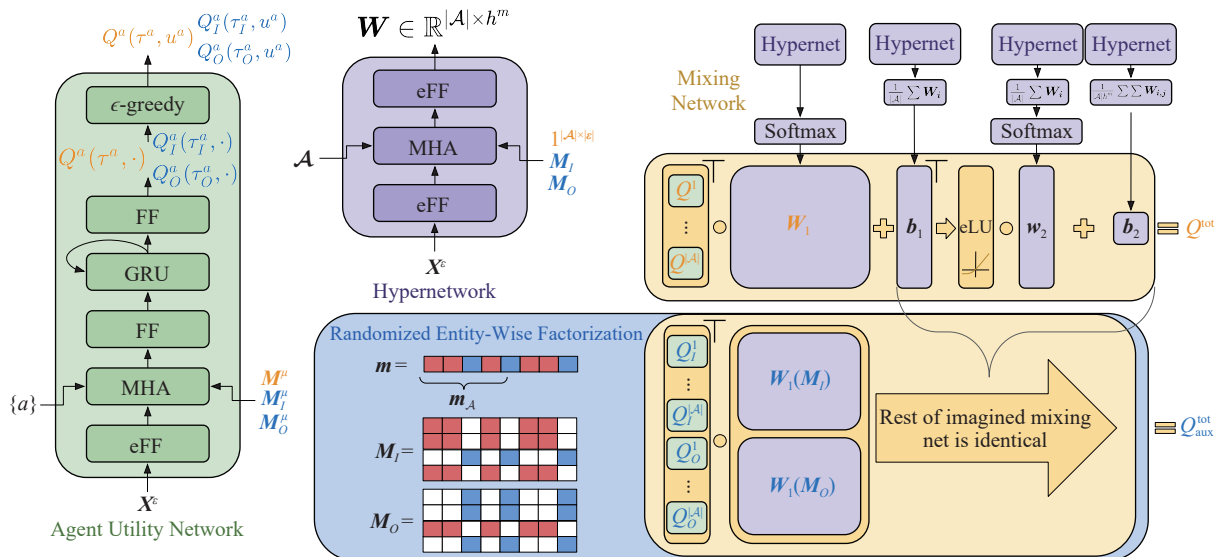


Fig. 6 Illustration of the REFIL Framework^[74]

PTM^[75] adopts REFIL's entity-wise framework and focuses on task prioritization in multi-task learning. It proposes a priority task discovery algorithm that emphasizes training on poorly performing tasks and enhances exploration by randomly masking agents' observation ranges. UPDeT^[79] aims to boost generalization performance *via* task prioritization and identification of similar substructures. OPT^[77] extends REFIL's idea of similar substructures across tasks, using MHA and GNN^[120] modules to extract structural information from multi-agent systems. It derives interaction prototypes to improve state representation when agent counts vary. HUANG *et al.*^[76] introduce a context-based meta-RL framework to enhance coordination in multi-agent teams with diverse and obscure attributes. By integrating context inference using historical data and refining it through contrastive learning, this method improves decision-making efficiency and precision across variable team compositions and capabilities. ALMA^[78] also employs an entity-wise framework, leveraging subtask similarities across tasks in a hierarchical decision-making structure. A high-level module assigns subtasks to agents, while a low-level module controls agents to execute those subtasks. Agents reduce their observation space based on assigned subtask vectors to focus on execution. However, in experiments, ALMA uses StarCraft II micromanagement scenarios, treating each enemy unit kill as a subtask, making the subtask assignment somewhat nominal. TETQmix^[80] also adopts the entity-wise framework, but effectively addresses the reward-related multi-task MARL challenges by integrating pretrained language models and a task-entity Transformer for efficient control and decision-making across varying tasks. It demonstrates superior learning efficiency and promising transferability to unseen tasks.

3.4 Multi-task Cooperation *via* Policy Distillation

OMIDSHAFIE *et al.*^[81] extend the single-agent policy distillation-based multi-task RL algorithm to the multi-agent domain, representing an early work in the field of multi-agent multi-task learning. They first train joint multi-agent policies for single-task scenarios and then distill the policies of agents across multiple tasks into a shared multi-task policy through supervised training. The distilled policy achieves performance comparable to that of single-task training in multi-agent systems of varying scales. Although this work makes some progress, subsequent research rarely built upon it to address multi-agent multi-task problems. SAML^[82] also adopts an offline distillation approach, following the technical route of the actor distillation algorithm in single-agent settings. It combines online training with offline distillation to achieve multi-task decision-making under varying numbers of agents.

During online training, separate MARL training is conducted for each task, while in the subsequent offline distillation phase, these pre-trained single-task policies are distilled into an integrated multi-task policy *via* supervised learning. Ref.^[121] proposes the double distillation network (DDN), which uses external and internal distillation modules to transfer global knowledge to local policies and enhance exploration *via* intrinsic rewards. Ref.^[122] distills agent-personalized global information into local knowledge, enabling efficient multi-task collaboration and decentralized execution. Many other algorithms explore policy transfer but do not adopt multi-task training or decision-making. Instead, they first train single-task policies on certain tasks, retain part of the policy modules for new tasks, and improve training efficiency on new tasks through fine-tuning.

3.5 Multi-task Cooperation *via* Policy Transfer

MATTER^[83] extends the QMIX framework based on an attention mechanism, structurally addressing the impact of varying agent numbers on decision-making and training. Its core idea is that efficient task representation could enhance agents' multi-task decision-making. To this end, MATTER separates the training and transfer phases. During training, the algorithm first trains task representations, using sampled data to predict the next state and reward signal based on the current state-action pair, then optimizes agent policies using MARL algorithms like QMIX combined with task representations. During transfer, the agent policy is fixed, and only the task representation is fine-tuned for unseen tasks. MRA^[84] extends meta-RL to multi-agent games with varying numbers of agents, constructing multiple submodules for task representation inference. During training, MRA samples low-level task features from a low-level representation module, combines them with agent observations to obtain high-level task features, and makes decisions based on a policy module. All modules are optimized using interaction data. For adaptation to unseen tasks, agents still sample low-level task features but only update the high-level module and policy module during training. PAT^[85] focuses on knowledge transfer in multi-agent scenarios, assigning each agent a student mode and a self-learning mode. Agents adjust between these modes based on their state. When unfamiliar with the environment, agents acquire knowledge from other agents *via* attention mechanisms for decision-making. When transferring to new tasks, the attention modules from old tasks are retained. MATE^[86] adopts an encoder-decoder framework to learn multi-agent task representations, using auxiliary tasks to provide state transition and reward function information for task representations. However, MATE's definition of multi-tasks mainly involves scenario size rather than differences in

agent numbers or reward functions, and testing on new tasks still requires some fine-tuning. VACL^[87] proposes an automatic curriculum learning-based policy transfer method to address multi-task multi-agent decision-making under sparse rewards. The authors claim that tasks with more units are more difficult. By modeling curriculum learning as an optimization problem, they gradually expand the training task set to improve the policy's ability to solve harder tasks.

3.6 Multi-task Cooperation via Modular Structure

In reward-related multi-task settings, it is challenging to use the shared backbone network to handle all tasks due to conflicting objectives among the tasks^[123]. Consequently, modular architectures, exemplified by the mixture of experts (MoE)^[124], have been widely adopted. Shared modular policies (SMPs) are proposed in Ref.^[90] to enable universal control policies by applying a single modular neural network to the actuators of morphologically diverse agents. By leveraging module sharing and local communication, SMP demonstrates strong generalization capabilities across various tasks. M3^[91] builds a modular policy representation using the MoE framework, and uses vector quantized variational autoencoder^[125] to discretize the latent space and learn transferable policy representations from offline multi-task, multi-agent data. It introduces task context embeddings and a latent space decomposition strategy, enabling efficient few-shot and zero-shot generalization through offline pretraining and online finetuning. ADMN^[92] assigns each agent a dedicated module and enables interaction *via* a dynamically learned graph based on observations, supporting flexible and scalable coordination in MARL. CSAOT^[93] introduces a mixture of policies (MoP) mechanism, decomposing the policy into sub-policies and using a gating network to select expert combinations based on task embeddings. It uses a top-k expert selection and weighted aggregation to improve adaptability, robustness, and inference efficiency in complex scenarios. YANHUI^[94] employs the MoE framework in multi-agent policies, where a gating function dynamically assigns observations to expert networks to enhance generalization. This structure shows strong performance in offline pretraining, particularly when training data quality is heterogeneous.

3.7 Multi-task Cooperation in Offline MARL

Given the higher cost of real-world multi-agent task data, research on offline multi-task MARL algorithms is crucial. Currently, MAGENTA^[88] and ODIS^[63] have extended single-agent offline multi-task RL algorithms to multi-agent scenarios. MAGENTA^[88] tackles multi-agent multi-task problems with differing reward functions, achieving multi-task decision-making in Honor of Kings,

Neural MMO, and StarCraft II micromanagement tasks. It employs different multi-task model setups, providing a Transformer-based state representation module for each task. These modules handle varying input space dimensions but have entirely different network structures and parameters for different tasks. The training utilizes a curriculum-based approach, separately training state representation modules and subsequent decision modules, resulting in three state representation modules (for three task types) and one shared decision module. Although MAGENTA obtains a strong multi-task multi-agent policy, it does not fully unify the multi-task model, limiting scalability. ODIS^[63] also proposes an offline multi-task MARL algorithm, with the core idea of leveraging similarities in cooperative behaviors across tasks to enhance agents' multi-task decision-making. ODIS divides policy training into two phases. Ref.^[126] discovers skills by reconstructing the next observations and uses behavior-regularized conservative Q-learning to perform generalization actions. INS^[89] uses interaction-aware synthesis *via* diffusion models to address data scarcity in offline MARL. By employing sparse attention and a bit action module, INS captures inter-agent dynamics and supports both discrete and continuous actions, leading to high-quality synthetic datasets that improve policy performance using minimal original data.

3.8 LLM-Based Multi-agent Cooperation

Cooperative MARL algorithms based on traditional neural network architectures struggle to achieve efficient cooperative decision-making in multi-domain and multi-task scenarios due to limitations in state and action feature representations. Currently, MAGENTA is also constrained by the complexity of multi-agent cooperation, requiring separate representation modules for each task. In contrast, LLMs^[127], which undergo extensive pre-training, post-training, and reinforcement alignment, have demonstrated superior instruction-following and decision-making capabilities^[128]. Consequently, to address multi-task cooperation in multi-agent systems, researchers have employed LLMs to construct cooperative multi-agent systems, enabling cooperative decision-making across multi-domain and multi-task scenarios using natural language^[129].

Several studies have focused on leveraging LLMs' summarization and decision-making abilities to transform traditional MARL testing environments into formats describable *via* natural language^[130]. Examples include StarCraft II^[97], complex cooperative tasks in embodied scenarios^[95,110], multi-player cooperative tasks in Minecraft^[131], cooperative and competitive games^[132-133], and grand-strategy diplomacy games^[96,134]. Embodied intelligence refers to the ability of agents to perform decision-making and control tasks while physically interacting with

complex and dynamic environments. For instance, TextStarCraft-II^[97] extracts multi-frame observation signals of all units from the game engine and converts them into text *via* scripts for storage in a queue, while using rule-based methods to filter out irrelevant information. Additionally, TextStarCraft-II introduces single-frame and multi-frame summarization techniques to enhance the flexibility of natural language-based cooperative decision-making. Researchers have also developed new testing environments featuring more challenging cooperative and competitive tasks framed in natural language, such as Werewolf^[135], multi-agent GitHub issue resolution^[98], and multi-robot cooperation^[99].

Some studies aim to propose more general algorithmic frameworks. As shown in Fig. 7, AutoGen^[100] leverages LLMs' universal comprehension and decision-making capabilities to solve broader cooperative problems, such as courtroom simulations^[136] and software/program development projects^[101]. AgentVerse^[137] assigns distinct roles and responsibilities to each agent for multi-stage cooperative decision-making, while CAMEL^[102] employs role-playing to enable multiple agents to autonomously collaborate with minimal human intervention. Agents are endowed with role-specific expertise and solve problems through dialogue and cooperation, accomplishing diverse tasks in an AI society and enabling multi-agent-based code generation. Approaches like RoCo^[107] and ChatEval^[106], along with related methods^[103-105], provide natural language-based communication mechanisms for agents, enhancing cooperative capabilities through discussions or even debates. MLAQ^[108] combines RL and LLM to develop a novel framework to generate virtual trajectories without interacting with the real environment. It retains the optimal decision-making ability of RL and the multi-task decision-making ability of LLMs, achieving state-of-the-art performance on multiple difficult problems. CoELA^[109] allows agents to autonomously determine communication timing and recipients for more efficient cooperation. Ref.[110] employs theory of mind (ToM)-based reasoning to achieve multi-agent cooperation across diverse textual tasks, using explicit belief states to mitigate LLM hallucinations.

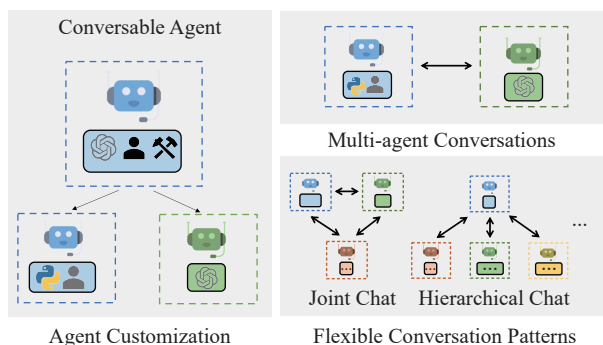


Fig. 7 Illustration of the AutoGen Framework^[100]

ProAgent^[111] constructs a proactive multi-agent cooperation framework comprising a planner, controller, verifier, cooperative error corrector, and memory module, dynamically adapting agent decisions to enhance teamwork. Although these works leverage the universality of natural language to achieve multi-task cooperative decision-making in complex scenarios, they do not utilize RL algorithms to optimize decision-making, remaining limited to formal multi-task implementations.

4 Benchmarks for Multi-task Cooperation

4.1 Scalability-Related Multi-task Cooperation Benchmarks

SMAC is a highly representative and challenging benchmark environment in the field of MARL. It requires the algorithm-controlled player to operate a group of agents to fight against enemies, incorporating both cooperation and competitive elements^[138-139]. The task is partially observable, where each agent receives observations of all relevant information, including distance, relative coordinates, health, shield, and unit type. The global state, on the other hand, includes the above information of all living units on the map, unrestricted by the field of view.

Each agent's action is divided into two parts. The environment-facing part includes stop and movement actions in four directions (up, down, left, right). The unit-facing part includes actions executable on targets, such as attack or heal. In addition, the number of agents dynamically changes as the battle progresses, so the action space of an individual agent also changes accordingly (*e.g.*, the dimension of attack or heal actions depends on the number of enemy or allied units).

REFIL further introduces an entity-centric multi-agent cooperation framework. The proposed variant of SMAC offers a scalability-related multi-task cooperation benchmark^[74]. In each scenario, the size of the initial multi-agent system is uncertain. For instance, in the 3-8 m scenario, which is a symmetric setting, the number of agents on both sides is randomly chosen from 3-8 at the beginning of each episode. This type of multi-agent cooperation environment better reflects the multi-task cooperation ability of a multi-agent system. Other similar scenarios include 3-8 Stalkers and Zealots and 3-8 Medivac, Marauders and Marines. Furthermore, some follow-up research has made improvements on this basis by constructing asymmetric competitive scenarios, for example, by maintaining randomness while always making the enemy team have one more unit than the player's team. Such settings are more capable of evaluating the cooperation performance of the algorithm.

4.2 Reward-Related Multi-task Cooperation Benchmarks

Since current cooperative MARL research lacks clearly defined multitask cooperative environments, this subsection constructs cooperative multi-agent environments with reward-related multi-task characteristics based on MA-Mujoco and Bi-DexHands (the code is available at <https://github.com/zhaozjie2022/reward-related-mtmarl/>). Specifically, the agents' input consists of local observations and task labels, and they are required to use a single model to cooperate across different tasks to verify their reward-related multi-task decision-making capabilities.

1) Multi-task MA-Mujoco: MA-Mujoco^[140] is an extended version of Mujoco designed for multi-agent cooperation. It divides a Mujoco suite into multiple agents based on joints and parts. For example, the Cheetah suite can be divided into 2 agents (corresponding to the front leg and hind leg) or 6 agents (each corresponding to one joint). The divided agents can only observe the positions, velocities, and angles of their respective joints, and can only output torques to control those joints. The environment allows agents to cooperatively control robots to perform a series of tasks, such as controlling the Cheetah run or the Hopper to hop. Following the reward-design of deepmind control suite^[141], MA-Mujoco can be further extended into a multi-task version, where a robot is controlled to perform various tasks. For instance, tasks for the Cheetah include: run, run backward, jump, run-front

(using only the front leg), and run-back (using only the hind leg). Each task has different semantics and reward functions, which can measure the algorithm's adaptability to multi-task scenarios.

2) Multi-task Bi-DexHands: Bi-DexHands^[142] is a simulation platform for bimanual dexterous manipulation based on Isaac Gym. In this environment, each agent controls a ShadowHand, and the two hands collaborate to complete manipulation tasks. The observations for each agent include joint information (position, velocity, angle, *etc.*) for their corresponding ShadowHand, as well as information about the objects (*e.g.*, the position and angle of the door). However, agents cannot directly access the physical information of the other hand, which makes it a partially observable environment. The multi-task version of Bi-DexHands requires the use of the same model to complete various tasks, such as opening doors, closing doors, pouring water from a kettle, and opening scissors. Bi-DexHands features a high-dimensional observation and action space, with dimensions up to 229 and 26, respectively, making it a challenging cooperative multi-agent environment.

Unlike scalable multi-task setting, the number of agents, observation dimensions, and action dimensions in these environments generally do not change over time. The main challenges lie in the variations of semantics and reward functions between tasks. As visualized in Fig.8, the task set may include completely unrelated tasks (*e.g.*, running and jumping) or even opposite tasks (*e.g.*, running forward and running backward), which further tests the model's multi-task performance.

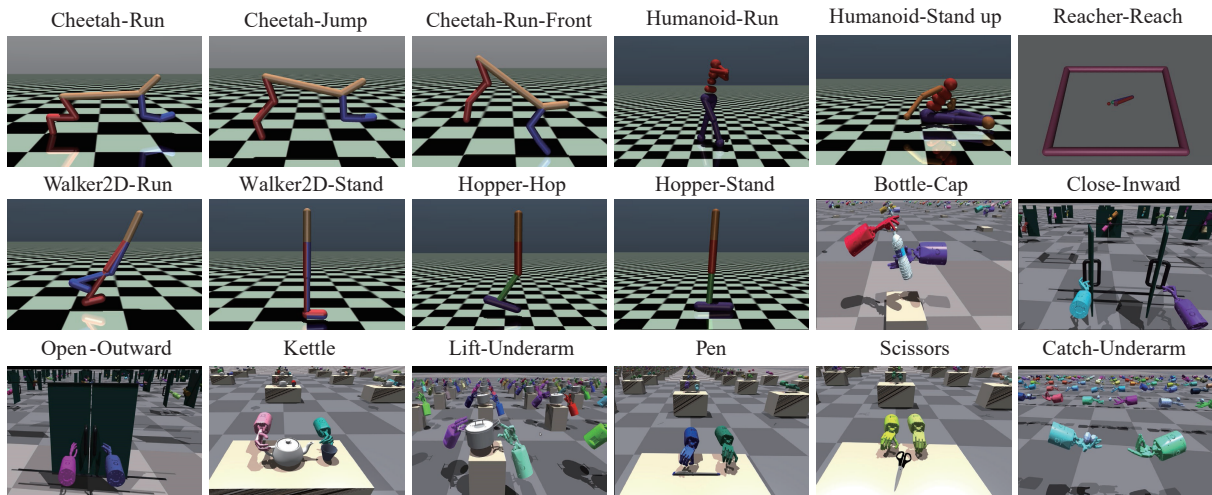


Fig. 8 Demonstrations of the Reward-Related Multi-task Multi-agent Environments

In the context of multi-task cooperative MARL, embodied platforms such as MA-Mujoco and Bi-DexHands provide natural testbeds that require agents to adapt to variations in dynamics, sensory feedback, and cooperative manipulation goals across different tasks. These platforms offer critical advantages, including realistic physical constraints, high-dimensional observation-action spaces, and rich

interaction modalities, making them ideal for studying multi-task cooperation. However, they also present notable challenges: the high complexity of the environments demands more sample-efficient learning algorithms, and partial observability, hardware noise, and delayed reward signals further complicate training and evaluation. Therefore, embodied intelligence platforms serve as both a per-

fect research framework and a stress test for developing robust multi-task MARL algorithms.

4.3 Mixed Multi-domain Multi-task Cooperation Benchmarks

Benefiting from the inherent flexibility of natural language, most existing multi-domain multi-task benchmarks are formulated as MDPs grounded in natural language. These benchmarks not only encompass scalability-related and reward-related multi-task challenges, but also introduce novel multi-domain challenges through variations in state and action spaces. LLMs possess strong generalization and transfer capabilities, allowing their abilities in task understanding, planning, prediction, and decision-making to be effectively adapted to different reward functions and task domains. In natural language-based multi-task cooperation benchmarks, some are adapted from existing cooperative benchmarks by converting the original MDPs into natural language formats, while others are specifically designed for decision-making in natural language.

In cooperative MARL algorithms, the StarCraft II micromanagement environment is the most prominent testing benchmark, with sufficient difficulty and complexity to validate MARL effectiveness. In the LLM era, TextStarCraft-II^[97] extracts multi-frame observation signals of all units from the game engine and transforms them into textual format *via* scripts for queue storage, using rule-based filtering to remove unimportant information. In addition, TextStarCraft-II proposes single-frame and multi-frame summarization methods to enhance the flexibility of natural language-based cooperative decision-making. Overcooked^[143] has also been modified in some studies to support natural language decision-making, where the environment describes each agent's state, task goals, environmental information, and additional details such as ingredient preparation time in natural language. However, the benchmarks used in these studies are often informal and lack consistency in prompt settings. In contrast, Collab-Overcooked^[144] constructs a dedicated natural language version of Overcooked^[145], containing 30 tasks across 6 complexity levels, with more comprehensive and consistent descriptions of agent states and environmental information. It also provides explicit communication mechanisms between agents to enhance cooperation. MindAgent builds upon Overcooked to create a 3D benchmark named Cuisineworld, in which agents must cooperate within a limited time to prepare and cook dishes^[146]. This benchmark is characterized by an increased number of agents and improved human-AI interaction, making it suitable for testing algorithms under higher task complexity. Welfare Diplomacy^[134] is based on the game Diplomacy^[147] and is

fully designed for natural language decision-making, where special rules and mechanisms guide players to pursue military expansion while also considering the optimization of domestic welfare. VillagerBench^[131] is a multi-agent cooperation benchmark built within the Minecraft^[148] environment. It tests multiple aspects of multi-agent cooperation through diverse task designs, including workload distribution, dynamic adaptation, and synchronized task execution.

Other benchmarks focus on multi-task cooperation for robotic platforms, aiming to leverage LLMs' general capabilities to empower embodied AI. TDW-MAT is based on ThreeDWorld platform^[149]. It develops a multi-agent multi-task scenario, which focuses on vision-guided tasks and motion planning. In TDW-MAT, Agents are required to transport scattered objects in simulated home environments and place them in designated locations. C-WAH^[150] is based on the multi-agent simulation environment of VirtualHome^[151], emphasizing cooperation and inter-agent communication. It supports two observation modes: symbolic and visual, so that the agents can perceive the environment through either structured object-level data or first-person RGB and depth visual inputs. RoCo-Bench offers a multi-robot arm cooperation benchmark, comprising six task domains to evaluate LLM-based multi-arm cooperation^[107]. Its codebase also provides a Sim2Real transfer suite, enabling direct interaction with physical robot arms for real-world deployment.

In addition, some benchmarks consider both cooperation and competition: BattleAgentBench^[152] evaluates agents' navigation, task execution, cooperation, and competition across seven sub-stages of varying difficulty, providing a comprehensive tool to assess LLMs in multi-agent settings. Multi-agent-Bench^[153] evaluates LLM-based multi-agent systems in diverse interactive scenarios. It could assess both cooperation and competition quality *via* key performance indicators (KPIs). The authors also examine various inter-agent communication protocols and agents' cooperation strategies.

4.4 Common Evaluation Protocols for Multi-task Cooperation

In evaluating multi-task multi-agent cooperation benchmarks, several commonly used metrics are adopted to assess algorithm performance. Key evaluation criteria include: ① Zero-shot performance, which measures how well agents perform on unseen tasks without further training; ② Task success rate, which computes the proportion of successfully completed tasks across different task settings; and ③ Policy transferability, which evaluates the agent's ability to adapt policies learned from one task to new tasks. In addition, when the environment supports communication or language-based cooperation, metrics like communication efficiency and coordination score are also employed.

These evaluation protocols ensure a comprehensive assessment of the agents' generalization, adaptability, and cooperation capabilities across varying

tasks and domains.

The above multi-task benchmarks and evaluation metrics are summarized in Table 4.

Table 4 A Review of Multi-agent Cooperation Benchmarks for Multi-task Scenarios

Benchmark	Category	Input State	Typical Tasks	Baselines	Metrics
Modified SMAC ^[74]	Scalability	Tensor	3-8_CSZ, 3-8_SYM_MM	REFIL ^[74] , OPT ^[77]	Win Rate
Multi-task Bi-DexHands	Reward	Tensor	Catch-Abreast/Underarm, Door-Open-Inward/Outward	MACD ^[48]	Average Return
Multi-task MA-Mujoco	Reward	Tensor	Cheetah-Jump/Run, Hopper-Hop/Stand	MACD ^[48]	Average Return
TextStarCraft-II ^[97]	Scalability	Language	StarCraft-II	SC-Phi2 ^[154]	Win Rate, Technology Rate
Collab-Overcooked ^[144]	Reward	Language	Cramped Room, Coordination Ring	Deepseek-v3 ^[155] , GPT-4o ^[156]	Progress Completeness, Initiating Capability, Responding Capability
Cuisineworld ^[146]	Reward	Image, Language	Salmon Meatcake, Lamb Meatcake	MindAgent ^[146]	Collaboration Score
VillagerBench ^[131]	Reward	Language	Workload Distribution, Synchronized Execution	VillagerAgent ^[131] , ProAgent ^[111]	Success Rate, Speedup
Welfare Diplomacy ^[134]	Reward	Language	Military Expansion, Domestic Welfare	Welfare Diplomacy ^[134]	Nash Welfare, Exploitability of Policy
TDW-MAT ^[149]	Reward	Image, Language	Food-Transporting, Stuff-Transporting	CoELA ^[109]	Transport Rate
C-WAH ^[150]	Reward	Image, Language	Prepare Afternoon Tea, Wash Dishes, Prepare a Meal	CoELA ^[109]	Success Rate, Speedup, Cumulative Reward
RoCo-Bench ^[107]	Reward, Multi-domain	Language	Sort Cubes, Pack Grocery, Sweep Floor	RoCo ^[107] , MLAQ ^[108]	Success Rate, Environment Steps
BattleAgentBench ^[152]	Multi-domain	Language	Navigation, Watch-and-Help	Qwen2-7B ^[157]	Forward Distance, Format Accuracy, Move Accuracy
Multi-agentBench ^[153]	Multi-domain	Language	Research, Coding, Minecraft	Llama3.3-70B ^[158]	Task Score

Note: CSZ: Colossus, Stalkers and Zealots; SYM_MM: Symmetric_marines vs marines; SC-Phi2: StarCraft-Phi2.

5 Challenges and Future Trends

The core objective of research in RL, particularly in multi-task cooperative MARL, is to address the decision-making challenges of real-world scenarios. This enables multi-agent systems in physical environments to achieve efficient cooperation under more complex task settings while integrating knowledge from different domains for effective policy transfer and generalization. Future research may focus on enhancing the applicability of algorithms in more complex and practical scenarios. This paper outlines potential future directions in the field from the following perspectives.

5.1 Multi-task Decision-Making in Cooperative-Competitive Games

Pure cooperative MARL algorithms primarily focus on controllable agents, achieving efficient cooperation within the system through methods like value decomposition^[19]. However, real-world scenarios often involve competitive elements, forming more intricate cooperative-competitive game problems. Despite existing efforts on cooperative-competitive problems^[159-161], research on multi-task scenarios remains limited. Future research could explore multi-agent problems with cooperative-competitive dynamics. Although a comprehensive

theoretical framework for general-sum games remains elusive, deep learning approaches may still enable multi-task cooperative-competitive decision-making in specific problem types, thereby expanding the applicability of related algorithms.

5.2 Multi-modal Multi-task Cooperation

Traditional cooperative MARL algorithms predominantly use vector-based state inputs, with only a few studies supporting image-based decision-making^[162]. While vision-augmented LLMs have endowed agents with multi-modal decision-making capabilities^[163], their ability to process video or audio modalities remains limited, despite these modalities being critical in real-world multi-agent scenarios. Future research could investigate multi-task cooperative decision-making under more diverse and complex modalities, leveraging existing algorithms to enhance real-world applicability.

5.3 Sim-to-Real Multi-task Cooperative Policy Transfer

The study of discrepancies between simulated and real-world environments is a widely recognized and important research direction. Addressing this challenge can facilitate the rapid transfer of multi-task cooperative policies trained in simulation to real-world settings^[164-165], reducing training costs and

improving practical cooperation performance. Future work should prioritize real-world multi-agent communication, proposing more efficient MARL algorithms that account for practical constraints (e.g., bandwidth, communication range, and frequency). Such advancements could solve real-world multi-task cooperation problems, such as multi-vehicle autonomous driving.

Furthermore, multi-task decision-making in real-world scenarios also faces the issue of high partial observability. The local observations obtained by agents may be delayed and severely incomplete compared to the global state, especially in large-scale multi-agent cooperative scenarios involving multiple tasks, where the difficulty of decision-making for agents will significantly increase.

6 Conclusion

This paper systematically reviews cooperative MARL for multi-task scenarios. First, we explicitly present three branches of multi-task problems under the multi-agent setting: scalability-related, reward-related, and multi-domain multi-task. We provide clear problem definitions and theoretical backgrounds from single-agent RL to multi-agent cooperation and further to multi-task multi-agent learning. Second, we comprehensively review cooperative MARL algorithms in both single-task and multi-task scenarios, and categorize these methods based on their technical approaches. Finally, we summarize the current benchmarks for multi-task cooperative MARL and present our insights on future research trends in this field.

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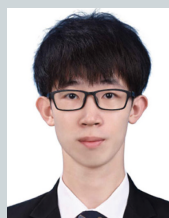
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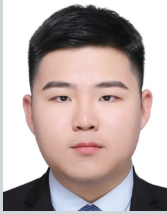
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